TEACHER GUIDE · GRADE 1 · UNPLUGGED SEQUENCING

# Match the Algorithm

Quick facilitation notes & answer key

## Goal

Students read two algorithms, figure out the job each one does, and match it — then write the algorithm for the leftover job. Reading code to determine what it accomplishes is abstraction and a core part of C3.2; writing the missing algorithm is C3.1.

## Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

## Materials

The printed worksheet and a pencil. Optional: cover the algorithm titles and have students guess the job from the steps alone.

## Run it (about 20 min)

1. I DO: read Algorithm 1's steps. 'Wet, soap, rinse, dry — these steps wash your hands.'
2. WE DO: Exercise 1 — Algorithm 1 = wash hands. Exercise 2 — Algorithm 2 (bowl, scoop, put down, call) = feed the dog.
3. YOU DO: Exercise 3 — the leftover job is 'get a drink'; students write its steps (e.g. get a cup, turn on the tap, fill the cup, turn off the tap).
4. Connect: 'Coders read code to figure out what it does — the steps tell the job.'

## Answer key (modelled by the run-gate)

Ex 1 — Algorithm 1 = wash hands (wet, add soap, rinse, dry).

Ex 2 — Algorithm 2 = feed the dog (get the bowl, scoop the food, put it down, call the dog).

Ex 3 — get a drink: a sample algorithm is get a cup, turn on the tap, fill the cup, turn off the tap. Accept any 3-4 sensible ordered steps.

## Watch for & differentiate

Common slip: matching by one word instead of the whole job — have students check that every step fits the job they chose.

Simplify: do Exercises 1 and 2 (matching) only.

Extend: ask students to write a 2-algorithm puzzle of their own for a partner to match.

Success indicator: the child matches Algorithm 1 to wash hands and Algorithm 2 to feed the dog, and writes a sensible 'get a drink' algorithm.